

22 Contract design

14 primary questions | Oldest to newest

2026 objectives: 3.5

21 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

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CP1-P2 April 2020 Q2	47 marks
CP1-P2 April 2021 Q1	42 marks

CA1-P2 | April 2005 | Q6

Current topic | 19 marks total | Original PDF p3

Related: Ch 33.

- 6** A company operates a benefit scheme. The benefit is a lump sum at the fixed retirement age, determined from salary and service in accordance with a set formula. At retirement, employees can choose either to take the lump sum as cash or to use it to purchase a pension from the insurance market.

- (i) State the information that would be useful for employees to receive every year about the plan and their benefit entitlement. [8]

The government is introducing legislation that requires retirement lump sums to be used to purchase a lifetime pension, rather than being taken as cash. Employee representatives have told the company that they are concerned that employees will be dependent on the insurance market. They suggest that the company offers an option for employees to convert their lump sum into an annual pension at an annuity rate of 35 for every 1000 lump sum, as an alternative to purchasing a pension from the insurance market. They comment that this will probably not cost anything as typical annuity rates in this country are currently 40 per 1,000 lump sum.

- (ii) Discuss the issues the company should consider in these circumstances. [11]
[Total 19]

CA1-P2 | April 2006 | Q8

Current topic | 22 marks total | Original PDF p4

Related: Ch 29, Ch 26.

- 8** An employer currently provides a wide range of employee benefits for its 10,000-strong workforce in a developed country. It is reviewing the effectiveness of its benefits programme and is considering introducing a flexible benefits system.
- (i) Explain how a flexible benefits scheme can be useful for an employer's corporate human resource strategy. [6]
 - (ii) Describe two examples of how employees could select against the employer in a flexible benefits system and for each suggest a way it could be mitigated. [3]

Company A is a general insurance company that writes an annual travel insurance policy with the following features:

- worldwide cover is provided for any number of trips during the year
- medical expenses abroad are covered up to \$10,000,000 per person
- repatriation costs are covered in the event of an accident while abroad
- the policy cover excludes business travel

Company A also writes single-trip travel insurance policies, normally purchased shortly before the trip starts.

Company A has suggested to the employer that an annual travel insurance policy, with monthly premiums deducted from salary, could be offered as an option in the flexible benefits system.

- (iii) Comment on how the risk profile of the policies that Company A would write under this arrangement might differ from the risk profile of its existing individual travel insurance portfolio. [5]
- (iv) List reasons why Company A may seek to purchase reinsurance. [5]
- (v) Outline the factors Company A will take into account when determining whether to purchase reinsurance to manage the risk on its travel insurance portfolio. [3]

[Total 22]

CA1-P2 | September 2006 | Q3

Current topic | 14 marks total | Original PDF p2

Examiner report and original paper: use the links for this question in [index.html](#).

3 An employer sponsors a final salary benefit scheme. The scheme includes the following features:

- Employees who leave employment with less than 3 years' service receive no benefit. Employees who leave after 3 years are entitled to a deferred pension payable from normal retirement age of 65 — deferred pensions are not increased between leaving and retirement.
- On death in service, the scheme pays a lump sum of 3 times salary to an employee's dependants. No benefits are payable on death in deferment.

The employer wants to amend these features of the scheme to make it more attractive to potential new recruits. His objective is to do this without significantly increasing the cost of the scheme to the employer.

- (i) Discuss why the existing scheme design may not appear attractive to individuals who are being recruited. [3]
 - (ii) Describe two changes to the benefit structure that might achieve the employer's objective, and explain how each change would achieve this objective. [4]
 - (iii) Discuss the implications for existing scheme members if the employer amends the scheme in any way that makes it more attractive to potential new recruits without significantly increasing the cost of the scheme to the employer. [7]
- [Total 14]

CA1-P2 | April 2008 | Q4

Current topic | 14 marks total | Original PDF p2

Related: Ch 23, Ch 7.

- 4** An insurance company is considering introducing a wedding insurance policy. The insurer intends to sell the policy through a large supermarket chain. The policy would be branded in the name of the supermarket, but the insurance company would cover the risks.
- (i) List the main risks that the policy should cover. [4]
 - (ii) Explain why the insurer may wish to use the supermarket as its sales channel. [4]
 - (iii) Discuss the rating factors the insurer could use in setting premiums. [6]
- [Total 14]

CA1-P2 | September 2010 | Q6

Current topic | 37 marks total | Original PDF p4

Related: Ch 26, Ch 33, Ch 34.

- 6** An industrial company employs 500 people. It currently operates a defined benefits pension scheme for its employees.

The company has decided to close this scheme for future service. From a given date, existing employees (and any subsequent new employees) will be invited to join a new defined contribution pension scheme. They will contribute the same amount to the new scheme as they do currently (or would have done) to the existing scheme. The employer will also contribute to the new scheme. Benefits in respect of service prior to the closing date will continue to be paid from the existing scheme under the rules and terms and conditions currently in force.

- (i) Describe the potential risks faced by the employer in providing a defined benefits scheme, which could have influenced the decision to close the current scheme. [8]
- (ii) Discuss why the proposed changes might cause industrial relations problems. [7]
- (iii) Set out the contents of the documentation that the company would be expected to provide to employees in relation to the new scheme. [5]
- (iv) Outline the considerations that should be taken into account when preparing this documentation and any other ongoing communications with employees. [7]

The contributions payable into the new scheme are to be invested in the units of one of five managed funds provided by an insurance company. Each employee can choose which one of the five funds “their” contributions will be allocated to. In practice, most employees opt for the domestic equity fund.

Five years after the commencement of the new scheme, the unit price of the domestic equity fund has fallen by 25%.

Representatives of the employees have approached the employer’s management to express their concerns. They argue that the fall in the unit price means that many employees are receiving or can expect to receive benefits that are considerably lower than those they were led to believe would be provided. To make up the shortfall, the representatives want the employer to make an additional one-off contribution and/or to increase future contributions payable in respect of all affected employees.

- (v) Discuss the points the employer should make when responding to this request.

[10]
[Total 37]

CA1-P2 | September 2011 | Q4

Current topic | 38 marks total | Original PDF p2

Related: Ch 5, Ch 25, Ch 8.

- 4** A retailer owns a chain of 15 stores selling high value, fashionable clothing and accessories. The chain's target markets are young, professional females. Most of the stores are located in and around the capital city of a developed country

- (i) Suggest why the profits of this retailer may be volatile. [3]
- (ii) Suggest adjustments that could be made to the retailer's business model that could reduce the volatility in profits. [3]

Following a run of several successful years, the retailer proposes to expand its operations significantly. This proposal will involve more than doubling the number of stores and the number of staff employed. The expansion will be restricted to the domestic country. The retailer has built its reputation on having highly knowledgeable and reliable staff, and is concerned that the expansion may compromise these standards.

In order to attract and retain the best staff and keep staff turnover as low as possible, the retailer has asked an insurance company to devise a savings scheme for employees that will encourage staff loyalty. This scheme will be in addition to the existing pay, bonus and benefits structure.

- (iii) Describe possible features of the savings scheme that could help to achieve the retailer's objectives. [7]

CA1-P2 | September 2011 | Q4 - continued

Current topic | 38 marks total | Original PDF p3

Related: Ch 5, Ch 25, Ch 8.

At present, the retailer does not provide pension arrangements for its employees. Employees are expected to rely on basic state pension provisions. The government has announced that it will introduce a new legal requirement for employers to make some provision for employees' pensions. The retailer is considering two alternative methods that will meet this requirement – neither of which will affect the employees' basic state provision:

- (a) Increase gross pay by a “pensions supplement”. Employees will be encouraged, but not required, to invest this supplement in a pension plan provided by an insurance company.
- (b) Both employer and employee contributions will be paid into a state top-up pension scheme. Contributions and the associated benefits are expressed as a proportion of basic salary and are set by the government.
- (iv) Discuss the advantages and disadvantages of each of these alternatives from the point of view of the employees. You do not need to consider any individual arrangements that some employees may already have. [11]

The retailer has identified that maternity and paternity provisions cause a significant cost and disruption to its planning. It is worried that the expansion will exacerbate these problems.

- (v) Outline measures the retailer could take to lessen the impact of these difficulties. You can assume that all reasonable actions would be legal. [5]

In order to finance the expansion, the retailer is proposing to issue a five-year conventional bond that will pay interest annually. Currently, yields on similar government bonds in the retailer's domestic country are 5% p.a. However, yields on such government bonds in a major overseas country are only 2% p.a. The retailer believes that by issuing the bond in the currency of this other country, it can cut its borrowing costs.

- (vi) Explain why yields on the respective government bonds may vary. [3]
 - (vii) Describe the risks faced by the retailer if it issued such a foreign currency bond. [6]
- [Total 38]

CA1-P2 | April 2013 | Q5

Current topic | 16 marks total | Original PDF p3

Related: Ch 18, Ch 23.

- 5** A large insurance company, which has been selling various personal lines products for many years, is now considering selling travel insurance.
- (i) Outline the advantages and disadvantages of:
- (a) selling the product directly
 - (b) using travel agents to sell the product
- [3]
- (ii) Outline the items of data that the insurance company would need in order to price this new travel insurance product. [6]
- The insurance company has invited an airline to act as a distribution channel for this product.
- (iii) Discuss the impact, on the insurance company's business model, of selling travel insurance through the airline. [7]
- [Total 16]

CA1-P2 | April 2014 | Q7

[Current topic](#) | [18 marks total](#) | [Original PDF p4](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 7** An insurance company is considering offering long-term care contracts. When an elderly person needs long-term care, the contract will meet the cost of care for the remainder of their life.
- (i) Discuss the factors that will need to be considered in determining a suitable design for this product. [15]
 - (ii) Outline how the answer to part (i) may differ if the contract had a cap on the total benefits provided. [3]
- [Total 18]

CA1-P2 | September 2016 | Q2

Current topic | 11 marks total | Original PDF p2

Related: Ch 19, Ch 0.

- 2** A defined benefit pension scheme provides, as the default option, a level pension to each member with a 50% spouse's pension on the death of the member.

However, the scheme allows members reaching retirement to choose any particular rate of payment escalation and any particular percentage spouse's pension.

The amount of the amended level of pension is then determined by the scheme applying adjustment factors to the default level of pension.

An actuary is advising the trustees of the pension scheme on what adjustment factors should be used, so as to minimise additional risks to the funding of the scheme under this arrangement.

- (i) Explain how the actuary should set the adjustment factors. [7]
- (ii) Outline how the actuary can use the actuarial control cycle in producing the advice. [4]

[Total 11]

CA1-P2 | September 2017 | Q7

Current topic | 25 marks total | Original PDF p5

Related: Ch 10, Ch 32.

7 For many years a large life insurance company has written substantial volumes of a flexible unit-linked investment product in a particular country. The product:

- provides a benefit of the full value of the units held on surrender at any time.
- allows the policyholder to vary their contributions.
- provides no additional benefit on death (other than the full value of the units).

The company is now planning to launch a guaranteed version of this product in a different country. All of the above product features will remain unchanged but a benefit guarantee will be added. The detailed terms and conditions of the guarantee are yet to be specified but the basic features are:

- at various points in time, units will be able to be cashed in for the higher of full value of the units and a minimum monetary amount.
 - the minimum monetary amount is based on an accumulation of contributions at guaranteed rates of interest set at outset.
- (i) Suggest possible reasons why the company may have decided to offer this guarantee. [5]
 - (ii) Consider a guaranteed contract that has just been issued and received its first contribution. Explain how the guarantee will have a value to the customer, immediately and in the future. [4]
 - (iii) Propose ways in which the detailed terms and conditions of the guarantee could be set to minimise the financial impact of the guarantee to the company. [9]
 - (iv) Explain why derivative investments are likely to be used to help the company manage the guarantee. [4]
 - (v) Discuss why derivative investments may not be completely effective in managing the risks associated with the guarantee. [3]

[Total 25]

CA1-P2 | April 2018 | Q7

Current topic | 19 marks total | Original PDF p3

Related: Ch 7, Ch 1.

- 7** A general insurance company is considering launching a cyber insurance policy aimed at small businesses.
- (i) Outline the cover that could be provided by this policy. [4]
 - (ii) Comment on how the needs and responsibilities of each of the relevant stakeholders might influence the design of the product. [4]
 - (iii) Discuss the factors the company needs to consider when designing the product. [5]
 - (iv) Comment on how your answer to part (iii) would change if the policy was aimed at:
 - (a) large companies.
 - (b) individuals.
- [6]
[Total 19]

CP1-P2 | April 2019 | Q2

Current topic | 50 marks total | Original PDF p4

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 2, Ch 3, Ch 26, Ch 35.

2 Background

The government of a developed country (Country B) is concerned about the low levels of personal savings by its citizens. The government is inviting a wide range of banks, insurance companies and investment companies to provide new and simple types of savings products to help meet its concerns.

The government is considering offering a range of incentives to help encourage increased savings.

The government wants the new types of savings products to:

- be low cost from the saver's point of view
- give the saver maximum flexibility on when they pay money into and take money out of the product.

An established investment company is considering launching a suitable product.

The company will administer this product as a unit-linked savings plan. There will be no death benefit. For this plan, the government will allow only one charge to be applied. The charge will take the form of an annual management charge levied on the unit fund, subject to a maximum of 0.75% of the fund value. The company wishes to make the product appeal to a wide range of potential savers.

The product will be aimed at all potential investors including those who might be less financially sophisticated.

The company needs to be able to compete with planned products of its competitors; some of these products are simple and non-linked.

Questions

- (i) Discuss why the government might want to increase the levels of personal savings in Country B. [7]
- (ii) Suggest, with reasons, possible encouragement and incentives that could be offered by the government to increase the levels of personal savings. [7]
- (iii) Discuss how the company's need to administer the product as a unit-linked plan, and yet be able to appeal to the complete range of financial sophistication of its target market, could be achieved by appropriate wordings of its marketing literature. [8]
- (iv) Discuss the advantages and disadvantages of the government's restriction on charges on the unit-linked plan, from the viewpoint of savers. [5]
- (v) Describe how and when the charges on the unit-linked plan could be disclosed to savers. [4]

CP1-P2 | April 2019 | Q2 - continued

Current topic | 50 marks total | Original PDF p5

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 2, Ch 3, Ch 26, Ch 35.

The company has now launched the savings product.

(vi) Discuss the possible risks to the company. [15]

A group of private investors is considering setting up a new company to provide the same type of unit-linked product as the established investment company.

(vii) Outline the main issues and opportunities that private investors would face, compared with the established investment company. [4]
[Total 50]

CP1-P2 | April 2020 | Q2

Current topic | 47 marks total | Original PDF p4

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 5, Ch 4, Ch 14, Ch 7, Ch 23, Ch 38.

2 Background

Country A is a developed country. In recent years there has been steady growth in the number of workers who are self-employed (i.e. people who work for themselves rather than being employed by a company) in Country A. The self-employed now make up 15% of the total workforce in Country A.

Research has shown that only 30% of the self-employed workers in Country A are actively saving for retirement. This compares with 75% of employed workers. The government in Country A has publicly expressed concern about the low levels of self-employed workers who are saving for retirement.

BBB is a large insurance company offering a wide range of products. It has seen sales and profits fall in all product lines over the last few years. BBB is considering a range of options to increase its future sales and profits.

One of the options being considered by BBB is to launch a simple unit-linked investment product. The product will allow individuals to invest in a range of unit-linked funds. BBB will target the product at self-employed workers with the aim of encouraging the self-employed to accumulate a fund during their working lives to save for their retirement. BBB believes that once an individual has bought the simple investment product, it will be able to sell other insurance products to the individual, which could further increase BBB's sales and profits.

Questions

- (i) Explain why the government of Country A is concerned about the number of self-employed workers not saving for retirement. [5]
- (ii) Discuss why the self-employed may not be saving for their retirement. [7]
- (iii) Propose possible actions the government could take to encourage self-employed workers to save adequately for their retirement. [5]
- (iv) Discuss the factors that will need to be considered in determining a suitable design for BBB's investment product. [9]
- (v) Discuss the types of investments that may be suitable for BBB's investment product. [5]
- (vi) List other possible insurance products that BBB might be able to sell to self-employed individuals. [5]
- (vii) Discuss the issues for BBB in pricing the products in part (vi). [7]
- (viii) Explain why it will be important for BBB to monitor the experience of this new business. [4]

[Total 47]

CP1-P2 | April 2021 | Q1

Current topic | 42 marks total | Original PDF p2

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 1, Ch 3, Ch 26, Ch 30, Ch 19.

1 Background

An insurance company (LifeCo) in a developed country with a high level of financial regulation is considering selling a new product providing long-term care benefits for individual citizens. LifeCo currently sells whole of life assurance and annuity products.

In recent years, the country's citizens have benefitted from significant improvements in life expectancy. Over the same period, there have been significant restrictions on government spending. This has led to the government reducing its financial support for meeting the costs of long-term care for its citizens. The government has indicated that it plans to cut its funding of long-term care further in future years.

In response to the government's proposed changes in long-term care funding, LifeCo intends launching a new long-term care product, which will be sold to people in employment. The product will offer a range of benefits that can be chosen by the individual. The benefits offered on the product may include:

- provision of nursing care in the policyholder's own home.
- cash payments to cover residential care.
- placements at purpose-built care homes, with different levels of care available depending on need.
- cover for funeral costs.

Competition for this type of product is currently very low with few competitors in the market. However, due to the reduction in government financial support for long-term care, a number of banks are also considering entering this market and offering each of the individual benefits listed above to their customers as standalone products.

Questions

- (i) List the advisers LifeCo may need to consult in setting up the new long-term care product. [3]
- (ii) Discuss the factors LifeCo should consider before deciding to launch this new product. [10]
- (iii) Describe the roles that the regulatory authorities may have in respect of the launch and ongoing management of LifeCo's new product. [5]
- (iv) Discuss the risks that LifeCo faces from other parties that could affect the success of LifeCo's new product. [8]
- (v) Discuss how LifeCo could mitigate the risks identified in part (iv). [6]

CP1-P2 | April 2021 | Q1 - continued

[Current topic](#) | [42 marks total](#) | [Original PDF p3](#)

Integrated case study. The full scenario is retained under one primary chapter; related chapters identify its other substantial topics. Related: Ch 1, Ch 3, Ch 26, Ch 30, Ch 19.

- (vi) Discuss the level and cover of benefits that LifeCo could offer to policyholders who have stopped paying premiums. [4]
 - (vii) Outline the assumptions LifeCo would need to set for pricing this product. [6]
- [Total 42]